

CT QC Copilot: Clinical Metrics Summary

Extracted from “Physics World Models for Computational Imaging”
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Overview

The CT QC Copilot translates the Triad Decomposition framework to clinical CT quality assurance. It implements ACR CT accreditation standards with automated metric computation, statistical process control for drift detection, and workflow efficiency optimization.

1 Gate Mapping: Research → Clinical CT

Table 1: Mapping of Triad Decomposition gates to clinical CT QC failure modes.

Gate	Research Interpretation	Clinical CT Interpretation
Gate 1	Information deficiency (null space, compression)	Protocol design inadequacy (insufficient projections, FOV truncation)
Gate 2	Carrier budget (SNR, photon count)	Dose budget (noise floor vs. diagnostic requirement)
Gate 3	Operator mismatch (mask shift, PSF error)	Scanner calibration drift (HU drift, CoR offset, gain variation)

Key finding: Gate 3 dominates in clinical practice—most QA failures trace to calibration drift, not protocol or dose issues. This is consistent with the research findings across all 7 validated modalities.

2 ACR Metric Validation

Nine ACR-aligned QC metrics computed automatically from CT phantom (Gammex 464) scans on a GE Revolution Apex scanner (120 kVp, 200 mAs, 5 mm axial).

Summary: ≤ 1.2 HU agreement for CT number accuracy; ≤ 0.15 mm for geometric accuracy. All metrics within $\leq 10\%$ of clinical tolerance bands.

3 Statistical Drift Detection (30-Scanner Fleet)

Statistical process control (SPC) using five Western Electric rules, evaluated on a 30-scanner simulated fleet over 12 monthly measurement cycles.

Table 2: ACR metric agreement: CT QC Copilot vs. console gold standard.

Metric	Copilot	Console	Deviation	% of Tolerance
CT# Water (HU)	0.3	0.2	0.1	2%
CT# Bone (HU)	952	953	1.0	5%
CT# Air (HU)	−997	−998	1.0	5%
CT# Acrylic (HU)	120	121	1.0	7%
CT# Polyethylene (HU)	−96	−96	0.0	0%
Geometric accuracy (mm)	0.15	0.10	0.05	2.5%
Slice thickness (mm)	5.08	5.02	0.06	4%
Uniformity (HU)	1.2	1.1	0.1	2%
Noise std. dev. (HU)	4.6	4.5	0.1	10%
Spatial resolution (lp/cm)	6.2	6.0	0.2	4%

3.1 Drift Scenarios Tested

- Noise drift: 2 scanners (gradual noise standard deviation increase)
- Uniformity drift: 1 scanner (progressive uniformity degradation)
- CT number drift: 1 scanner (systematic HU calibration shift)

3.2 Detection Performance

Table 3: Drift detection performance summary.

Metric	Value
Sensitivity	100% (4/4 drifting scanners)
95% CI (sensitivity)	39.8%–100%
Specificity	100% (0/26 false positives)
95% CI (specificity)	86.8%–100%
Early warning lead time	3–6 months before ACR threshold
False positive rate	0%

4 Workflow Efficiency

4.1 Annual Impact (30-Scanner Fleet)

5 Artifact Signatures

Six artifact signatures computed by the Copilot correspond directly to OperatorGraph mismatch parameter families:

6 CT Correction Results

From the main paper validation:

Table 4: Per-scanner QC workflow time comparison: manual physicist workflow vs. CT QC Copilot.

Stage	Manual (min)	Copilot (min)
DICOM ingestion	5.0	0.01
Metric computation	25.0	0.01
Threshold evaluation	5.0	<0.01
Trend analysis	10.0	<0.01
Report generation	15.0	0.02
Physicist review/sign-off	7.0	4.0
Total	67 $\pm$ 12	4.2 $\pm$ 0.8

Table 5: Annual workflow savings projection.

Metric	Value
Time reduction per scanner	94%
Physicist-hours saved annually	377 hours
FTE reallocation	0.18 FTE
Reproducibility	Bit-exact (SHA-256 verified)
Inter-analyst variability	Eliminated

Table 6: CT artifact signatures and their corresponding mismatch parameters.

Artifact	Mismatch Parameter	Clinical Impact
Ring artifacts	Center-of-rotation offset	Diagnostic confusion
Cupping artifacts	Beam hardening coefficient	CT# inaccuracy
Streak artifacts	Metal/motion	Obscured anatomy
HU drift	Detector gain calibration	Quantitative error
Noise ratio change	Tube current / filtration	Dose optimization
Geometric distortion	Gantry alignment	Treatment planning error

- **Mismatch type:** Sub-degree center-of-rotation offset
- **Scenario II PSNR:** 13.41 dB (severe ring artifacts)
- **Scenario III PSNR:** 24.09 dB (corrected)
- **Correction gain:** +10.68 dB
- **Solver:** FBP with Ram-Lak filter

Opportunities for Collaboration

1. **Prospective clinical validation:** Deploy CT QC Copilot on real clinical scanner fleet at UT Southwestern with physical ACR phantoms, replacing simulated fleet validation.
2. **MRI clinical validation:** Validate coil sensitivity mismatch correction on clinical MRI data (fastMRI + institutional data).
3. **Regulatory pathway:** Guide FDA 510(k) or De Novo pathway for CT QC Copilot as clinical decision support software.
4. **Multi-institution study:** Extend drift detection validation across institutions with diverse scanner manufacturers (GE, Siemens, Philips, Canon).